

MIR Medical Intelligence: Research Proposal and Integrated Research Roadmap

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Independent Research Program / MIR Medical Intelligence

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Abstract

This document provides the scientific and technical research proposal for MIR Medical Intelligence. The program investigates a missingness-aware, uncertainty-aware clinical intelligence framework that reasons over available laboratory and clinical evidence, communicates evidence coverage and uncertainty, identifies information gaps, and investigates whether additional evidence can be selected when it is expected to improve a clinical assessment. The initial research scope is restricted to non-imaging laboratory and structured clinical data, with later expansion toward multimodal medical intelligence. This proposal defines the research questions, hypotheses, datasets, methodology, model-development program, evaluation strategy, explainability and uncertainty framework, adaptive evidence-acquisition research track, federated learning and agentic-AI extensions, governance, safety, reproducibility, publication strategy, work packages, and multi-year roadmap.

Index Terms

medical artificial intelligence, clinical decision support, machine learning, deep learning, laboratory medicine, missing data, uncertainty quantification, explainable AI, adaptive evidence acquisition, federated learning, agentic AI, clinical intelligence

DOCUMENT CONTROL

TABLE I
PROPOSAL METADATA.

Version	v1.0
Date	September 2026
Project	MIR Medical Intelligence
Research leads	Mir AbdulRehman; Sababa Ateeq
Document type	Research proposal and integrated research roadmap
Status	Working research document; update version on substantive changes

I. EXECUTIVE SUMMARY

- A. Program objective*
- B. Primary deliverables*
- C. Expected contribution*
- D. Current research position*

II. RESEARCH TEAM, LEADERSHIP, AUTHORSHIP, AND PROVENANCE

- A. Research team and roles*
- B. Research leadership model*
- C. Research identity and persistent identifiers*
- D. Authorship and contributor policy*
- E. CRediT contribution tracking*
- F. Research provenance and project origin*
- G. External collaborators, supervisors, and institutional partners*
- H. AI-assisted research and disclosure policy*

III. BACKGROUND AND RESEARCH CONTEXT

- A. Clinical and scientific context*
- B. Problem domain*
- C. Current approaches and their limitations*
- D. Motivation for an evidence-aware framework*
- E. Preliminary rationale*

IV. PROBLEM STATEMENT

- A. Primary problem*
- B. Clinical and methodological gap*
- C. Operational constraints*
- D. Research assumptions*
- E. Out-of-scope claims and boundaries*

V. RESEARCH GAP AND NOVELTY

- A. Existing research landscape*
- B. Unmet methodological needs*
- C. Novel aspects of the proposed program*
- D. Expected scientific contribution*

- Supervised clinical-risk prediction.
- Longitudinal temporal modeling.
- Missingness-aware inference.
- Adaptive evidence acquisition.
- Explainability and uncertainty quantification.
- Federated learning.
- Safe agentic clinical intelligence.
- Future multimodal medical intelligence.

IX. SCIENTIFIC PREMISE AND SYSTEM OPERATING PRINCIPLE

- A. Evidence-in-hand first*
- B. Partial observability as a design requirement*
- C. Inference, uncertainty, and evidence coverage*
- D. Inconclusive outputs and abstention*
- E. No fabrication of unavailable observations*
- F. Prediction is not diagnosis*

X. DATASET STRATEGY AND ACQUISITION ROADMAP

- A. Dataset portfolio*
- B. Dataset inclusion criteria*
- C. Recommended acquisition order*
- D. Synthetic and open data*
- E. Credentialed clinical data*
- F. Longitudinal EHR research*
- G. Future imaging resources*
- H. Dataset provenance records*
- I. Data access, licenses, and use restrictions*

XI. STUDY DESIGN AND COHORT CONSTRUCTION

- A. Study design progression*
- B. Population definition*
- C. Index date and observation windows*
- D. Inclusion and exclusion criteria*
- E. Outcome definitions*
- F. Cohort construction*
- G. Temporal splitting*
- H. External validation design*
- I. Ascertainment and indication bias*
- J. Causal interpretation limits*

XII. DATA STANDARDIZATION AND CLINICAL DATA MODEL

- A. Canonical internal schema*
- B. Laboratory ontology*

XXIII. SOFTWARE ARCHITECTURE AND RESEARCH INFRASTRUCTURE

- A. Repository structure*
- B. Python environment*
- C. R companion environment*
- D. Experiment tracking*
- E. Configuration management*
- F. Continuous integration and testing*
- G. Artifact versioning*
- H. Model registry*

XXIV. RESEARCH WORK PACKAGES

- A. WP0 — Governance and provenance*
- B. WP1 — Laboratory data foundation*
- C. WP2 — Anomaly intelligence*
- D. WP3 — Supervised risk prediction*
- E. WP4 — Longitudinal modeling*
- F. WP5 — Missingness-aware inference*
- G. WP6 — Uncertainty and calibration*
- H. WP7 — Adaptive evidence acquisition*
- I. WP8 — External validation*
- J. WP9 — Federated learning*
- K. WP10 — Agentic intelligence*
- L. WP11 — Clinical interface and safety case*
- M. WP12 — Publication and dissemination*
- N. WP13 — Multimodal expansion research*

XXV. FIRST 90 DAYS — CONCRETE EXECUTION PLAN

- A. Days 1–15: scientific and data foundation*
- B. Days 16–30: laboratory representation*
- C. Days 31–60: anomaly research*
- D. Days 61–90: supervised baseline*

XXVI. 24-MONTH AND 36-MONTH INTEGRATED RESEARCH ROADMAP

- A. Phase 1 — Foundation*

XXVII. RESEARCH DECISION GATES

- A. Gate 1 — Data readiness*
- B. Gate 2 — Baseline reproducibility*
- C. Gate 3 — External validity*
- D. Gate 4 — Uncertainty reliability*
- E. Gate 5 — Adaptive-evidence feasibility*
- F. Gate 6 — Federated feasibility*
- G. Gate 7 — Agentic safety readiness*
- H. Gate 8 — Multimodal expansion readiness*

XXVIII. RESEARCH GOVERNANCE, ETHICS, PRIVACY, SAFETY, AND REGULATION

- A. Human-subject and data-governance principles*
- B. Data minimization and access controls*
- C. Anonymization and pseudonymization considerations*
- D. GDPR and applicable data-protection requirements*
- E. Medical-device and software implications*
- F. EU AI Act considerations*
- G. Risk management and ISO 14971 alignment*
- H. Clinical safety case*
- I. Hazard identification*
- J. Human oversight*
- K. No autonomous diagnosis or treatment policy*

XXIX. REPRODUCIBILITY AND OPEN-SCIENCE PLAN

- A. Repository structure*
- B. Dataset cards*
- C. Model cards*
- D. Research logs*
- E. Experiment registry*
- F. Version control*
- G. Archival releases and DOI strategy*
- H. Code and data-sharing boundaries*
- I. Reproducible environment*

XXXIII. SUCCESS CRITERIA

- A. Scientific success*
- B. Statistical success*
- C. Engineering success*
- D. Clinical-safety success*
- E. Reproducibility success*
- F. Publication success*

XXXIV. RESOURCES, INFRASTRUCTURE, AND BUDGET ASSUMPTIONS

- A. Compute*
- B. Software*
- C. Storage*
- D. Human expertise*
- E. External services*
- F. Expected costs and free-resource strategy*

XXXV. IMMEDIATE ACQUISITION AND SETUP CHECKLIST

- Persistent researcher identifiers (ORCID).
- Research repository and version control.
- Open research project workspace.
- Shared reference-management library.
- Governance and contribution files.
- Dataset access and provenance records.
- Study protocol.
- Experiment registry.

XXXVI. EXPECTED PUBLICATIONS AND RESEARCH OUTPUTS

- 1) Protocol or framework paper.
- 2) Laboratory anomaly and standardization study.
- 3) Longitudinal laboratory prediction study.
- 4) Missingness-aware and uncertainty-aware inference study.
- 5) Adaptive evidence-acquisition study.

- 6) Federated-learning study.
- 7) Safe agentic clinical-intelligence study.
- 8) Future multimodal study.

XXXVII. REFERENCES AND SOURCE REGISTRY

APPENDIX A

EXAMPLE RESEARCH LOG TEMPLATE

TABLE IV

MINIMAL EXPERIMENT LOG FIELDS.

Date
Researcher
Experiment ID
Research question
Dataset/version
Code commit
Configuration
Result summary
Limitations/issues
Next action

APPENDIX B

MINIMUM MODEL CARD TEMPLATE

- Model name and version.
- Intended use.
- Out-of-scope use.
- Training data.
- Evaluation data.
- Input variables and missingness behavior.
- Performance and calibration.
- Subgroup performance.
- Known limitations.
- Safety considerations.
- Version and release information.

APPENDIX C

RESEARCH POSITION STATEMENT

The project is a staged research program. Claims should increase only as evidence, validation, and reproducibility increase. Early outputs should be treated as research findings rather than clinical deployment claims.